A07: Independent mathematical audit of P07
Five-node golden-ratio specialization

Independent reconstruction and adversarial source review

3 October 2026

1 Verdicts and exact-version disposition

FOURIER / TRIGONOMETRY VERDICT: VERIFIED
ORDERING / SELECTION VERDICT: VERIFIED
FREQUENCY / COINCIDENCE VERDICT: VERIFIED
STABILITY / CRITICAL-MODE VERDICT: VERIFIED
SCOPE / DEPENDENCY VERDICT: VERIFIED
OVERALL VERDICT: VERIFIED

Every substantive assertion in the current P07 source passes under its stated positive canonical equilibrium assumptions. There is no required mathematical correction, no meaning-changing wording correction, and no unresolved mathematical step. In particular, the full numerical-coincidence classification and the strict stability of the *entire* dominant tie hold. The physical-parameter codimension claim has the necessary regular branch justification. These verdicts are based on the derivations below; finite checks and historical validation are separate corroboration.

Required corrections: none. This audit does not edit P07 or mark its project status VERIFIED. The source, registry, manifest and compendium retain their supplied DRAFT status. Audit closure belongs to the original P07 development chat. **P09 is cleared to begin after that closure is recorded**; it is next in the frozen attempt sequence, and its direct prerequisites P02/P03/P04 are already audited. No P09 or later proof is begun here.

The exact audited file is `proofs/P07_five_node_specialization.tex`, **P07 v1.0, 2026-10-03**, with **24,180 bytes and 555 lines**. All P07 line references in this audit count comments and blank lines in that exact source. Its full-file SHA-256 is

`fc10d257e8126b79a56cc7dd4c55b14b0266e250cbe40dbecb6c6a18658d09de`

and its mathematical-body SHA-256, from the first line-anchored `\section` command through EOF, is

`66a5b8f078cb0fc30f4d7c72c22a26b965fd8ded491f53450d604fe7a21ccdea`

Both equal the CURRENT P07 manifest record. The supplied `cyclic_network_project_P07_draft.zip` has SHA-256

`2a49a916659df343995adf5b6e75975f1d05b97f2d0bf24f43db631d0472106c`

and contains 118 files, all under `cyclic_network_project/`. Its CRCs pass. The other current delivery copy has identical archive bytes and identical P07 source bytes. Older local TODO placeholders were excluded; P07 was never reconstructed from conversation history.

2 Current inputs and dependency authentication

All 23 requested current inputs were located in the authenticated archive before the mathematical review. The manifest's final P07 entry supersedes historical scheduling prose. The supplied current P01–P05 and P08 proof full-file and body hashes all match their manifest closure records. The associated A01–A05/A08 audits are retained as audits of their original versions; their original verdicts are not rewritten by later closure.

Current input	Use in this audit
`project_manifest.md`	Exact source selection, current hashes and closures.
`claim_registry.md`	C051/C052 algebra and selection; rejected physiological C155; P07-D1–D6 mapped to source. No status edits.
`frozen_theorem_package.md`	Supporting five-node role, dependency discipline, release gate and attempt order.
`final_novelty_verdict.md`	Existing bounded priority assessment; no new priority claim or survey.
`novelty_decisions.md`	Supporting specialization and interpretation limits, read with the superseding frozen package.
`02_theorem_dependency_map.tex`	Direct P05 $\rightarrow$ P07 edge; P08-before-P07 is scheduling, not a theorem edge.
`proofs_compendium.tex`	Current lifecycle front matter and actual P07 input filename.
`shared_macros.tex`	Fourier, real-part, imaginary-part and theorem notation.
`00_canonical_model.tex`	Original five-term DM equation, positive physical domain and orientation.
`01_notation_and_assumptions.tex`	Zero-based labels, predecessor shift, unitary Fourier convention, selected-set and stability meanings.
P01 v1.1 and A01	Inherited smoothness and well-posedness foundation. A01's sole root-wording correction is closed.
P02 v1.1 and A02	Original synchronous restriction and positive-equilibrium bracket. The ordered-root wording correction is closed; the original A02 TeX is present despite the historical source-absence header.
P03 v1.1 and A03	Homogeneous coefficients and equilibrium simplification of d; audited body preserved.
P04 v1.1 and A04	Oriented spectrum, unitary diagonalization, semisimple multiplicities and strict linear stability test; audited body preserved.
P05 v1.1 and A05	Exact $q = 2$ cells, all-sign χ convention and local equilibrium chart. The candidate-set wording correction is closed.
P08 v1.1 and A08	Version/scope context only. Its corrected $n = 8$ witness label is present. No MB or P08 theorem is used.
P07 v1.0	The exact 555-line source being audited, not a prerequisite assumed true.

Full hashes, byte counts, line counts and proof-body hashes for every entry are in `A07_support/input_hashes.json`; the complete 118-member inventory is in `A07_support/archive_member_hashes.json`. This avoids confusing an original audited version with its current corrected closure. The A04/A05 historical validation-provenance qualifications remain historical; they do not weaken their accepted mathematical inputs or become P07 premises.

The specific imported results are P02's synchronous scalar restriction; P03 Proposition `prop:P03-homogeneous` and `eq:P03-equilibrium-d`; P04 Theorem `thm:P04-spectrum`, Proposition `prop:P04-multiplicity`, and Corollary `cor:P04-stability`; and P05 Theorem `thm:P05-selection` with `eq:P05-regions`, `eq:P05-equilibrium-chart` and `eq:P05-chart-determinant`. P05's chart argument is in current source lines 401–425. No external literature is needed for this audit.

3 Canonical setup and exact fifth-root reconstruction

Write $x = x^* > 0$ and $s = 1 + x$. The original synchronous DM component is

$$ax - kx^2 + \frac{cx}{s} - \frac{gx}{s} - hx^2 = x\left[a - (k+h)x + \frac{c-g}{s}\right].$$

Thus P07's bracket vanishes at its specified equilibrium. With $C = c/s$, $G = gx/s^2$, $H = hx$, the original diagonal derivative gives

$$\begin{aligned} d &= a - 2kx - \frac{cx}{s^2} - \frac{g}{s} - hx \\ &= (k+h)x - \frac{c-g}{s} - 2kx - \frac{cx}{s^2} - \frac{g}{s} - hx \\ &= -kx - C - \frac{Cx}{s}. \end{aligned}$$

This last identity requires equilibrium; the ordering calculations alone do not permit independent physical variation of d.

Let $\phi = (1+\sqrt{5})/2$. Squaring gives $\phi^2 = \phi + 1$; division by $\phi > 0$ gives $\phi^{-1} = \phi - 1$. Hence

$$\phi^{-2} = (\phi-1)^2 = 2 - \phi = \frac{3-\sqrt{5}}{2}, \qquad \phi + \phi^{-1} = 2\phi - 1 = \sqrt{5}.$$

For $z = e^{2\pi \mathrm{i}/5}$, the geometric sum yields $1 + 2u_1 + 2u_2 = 0$, where $u_1 = \cos(2\pi/5)$ and $u_2 = \cos(4\pi/5) = 2u_1^2 - 1$. Thus $4u_1^2 + 2u_1 - 1 = 0$. The angle $2\pi/5$ is in quadrant I, so the positive root must be selected:

$$u_1 = \frac{\sqrt{5}-1}{4} = \frac{1}{2\phi}, \qquad u_2 = -\frac{\sqrt{5}+1}{4} = -\frac{\phi}{2}, \qquad u_1 - u_2 = \frac{\sqrt{5}}{2}.$$

Since $\cos(\pi/5) = -u_2 = \phi/2$ and $\pi/5$ is in quadrant I,

$$\sigma := \sin(\pi/5) > 0, \qquad \sigma^2 = 1 - \frac{\phi^2}{4} = \frac{5-\sqrt{5}}{8}, \qquad \sigma = \frac{\sqrt{10-2\sqrt{5}}}{4}.$$

The double-angle formula gives $\sin(2\pi/5) = \phi\sigma > 0$. Its square is $(5+\sqrt{5})/8$, so positivity fixes the radical $\sqrt{10+2\sqrt{5}}/4$. No decimal comparison establishes any radical identity here.

Reducing $\theta_m = 2\pi m/5$ and $2\theta_m$ modulo 2π reconstructs the following table independently:

m	$\cos\theta_m$	$\sin\theta_m$	$\cos 2\theta_m$	$\sin 2\theta_m$
0	1	0	1	0
1	u_1	$\phi\sigma$	u_2	σ
2	u_2	σ	u_1	$-\phi\sigma$
3	u_2	$-\sigma$	u_1	$\phi\sigma$
4	u_1	$-\phi\sigma$	u_2	$-\sigma$

In particular, $2\theta_2 = 8\pi/5$ is in quadrant IV and $2\theta_3 = 12\pi/5$ reduces to quadrant I. Rows 4 and 3 are the conjugate rows of 1 and 2. There is no Nyquist singleton in odd dimension; the classes have Fourier labels $\{0\}$, $\{1,4\}$, and $\{2,3\}$.

4 Oriented spectrum and complete real-part ordering

Starting with verified P04, not with a P07 formula,

$$\begin{aligned}\lambda_m &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{2\mathrm{i}\theta_m} - He^{-2\mathrm{i}\theta_m} \\ &= d + C\cos\theta_m + (G-H)\cos 2\theta_m + \mathrm{i}[-C\sin\theta_m + (G+H)\sin 2\theta_m].\end{aligned}$$

The last shift contributes $-H\cos 2\theta_m + \mathrm{i}H\sin 2\theta_m$. Its imaginary sign is positive. Put $T = G - H$ and $S = G + H$; the phase table then gives exactly

$$\begin{aligned}\lambda_0 &= d + C + T, \\ \lambda_1 &= d + \frac{C}{2\phi} - \frac{\phi T}{2} + \mathrm{i}\sigma(S - \phi C), \\ \lambda_2 &= d - \frac{\phi C}{2} + \frac{T}{2\phi} - \mathrm{i}\sigma(C + \phi S), \\ \lambda_4 &= \overline{\lambda_1}, \qquad \lambda_3 = \overline{\lambda_2}.\end{aligned}$$

Thus the three real parts are

$$r_0 = d + C + T, \quad r_1 = d + \frac{C}{2\phi} - \frac{\phi T}{2}, \quad r_2 = d - \frac{\phi C}{2} + \frac{T}{2\phi}.$$

These are also $R(1), R(u_1), R(u_2)$ for P05's $R(u) = d - T + Cu + 2Tu^2$, because $2u_1^2 - 1 = u_2$ and $2u_2^2 - 1 = u_1$. The pair members share real parts without necessarily sharing numerical eigenvalues.

Subtracting before factorization gives

$$\begin{aligned}r_1 - r_2 &= \tfrac{1}{2}(\phi + \phi^{-1})(C - T), \\ r_0 - r_1 &= (1 - \tfrac{1}{2\phi})C + (1 + \tfrac{\phi}{2})T, \\ r_0 - r_2 &= (1 + \tfrac{\phi}{2})C + (1 - \tfrac{1}{2\phi})T.\end{aligned}$$

Since $1 + \phi/2 = \sqrt{5}\phi/2$ and $1 - 1/(2\phi) = \sqrt{5}/(2\phi)$, every claimed prefactor follows:

$$\begin{aligned}r_1 - r_2 &= \frac{\sqrt{5}C}{2}(1 - t), \\ r_0 - r_1 &= \frac{\sqrt{5}\phi C}{2}(t + \phi^{-2}), \\ r_0 - r_2 &= \frac{\sqrt{5}C}{2\phi}(t + \phi^2),\end{aligned} \tag{1}$$

where $t = T/C$ is well defined. Each prefactor is positive and $-\phi^2 < -\phi^{-2} < 1$. Reconstructing the ordering from these signs, in the order $(r_1 - r_2, r_0 - r_1, r_0 - r_2)$, yields

Condition	Gap signs	Ordering
$t > 1$	$(-,+,+)$	$r_0 > r_2 > r_1$
$t = 1$	$(0,+,+)$	$r_0 > r_1 = r_2$
$-\phi^{-2} < t < 1$	$(+,+,+)$	$r_0 > r_1 > r_2$
$t = -\phi^{-2}$	$(+,0,+)$	$r_0 = r_1 > r_2$
$-\phi^2 < t < -\phi^{-2}$	$(+,-,+)$	$r_1 > r_0 > r_2$
$t = -\phi^2$	$(+,-,0)$	$r_1 > r_0 = r_2$
$t < -\phi^2$	$(+,-,-)$	$r_1 > r_2 > r_0$

The rows partition the entire real line, including every threshold. The distinct gap zeros exclude a triple tie of *classes*. The later triple *numerical value* involves only classes 0 and 1, so there is no conflict. Every comparison cancels d.

5 Selection, midpoint consistency, and canonical switches

Pair 2 is excluded directly: for $t < 1$, $r_1 - r_2 > 0$; for $t \geq 1$, $r_0 - r_2 > 0$ since $t + \phi^2 > 0$. Hence it never maximizes, even at either subdominant equality. The remaining maximum is determined by the second gap in (1). With $D_5 = -T = H - G$,

Condition	Representatives	Fourier labels
$D_5/C < \phi^{-2}$	$\{0\}$	$\{0\}$
$D_5/C = \phi^{-2}$	$\{0,1\}$	$\{0,1,4\}$
$D_5/C > \phi^{-2}$	$\{1\}$	$\{1,4\}$

This conclusion is independently proved, not inferred only from P05's reachable-mode theorem.

The P05 midpoint values agree exactly:

$$\beta_0 = \frac{1+u_1}{2} = \frac{2+\phi^{-1}}{4} = \frac{\phi^2}{4}, \quad \beta_1 = \frac{\phi^{-1}-\phi}{4} = -\frac{1}{4}, \quad \alpha_0 = \frac{1}{\beta_0} = \frac{4}{\phi^2} = 6 - 2\sqrt{5}.$$

When $D_5 > 0$, its vertex $u_* = C/(4D_5) > 0$ reaches the positive midpoint β_0 exactly at $D_5 = C/\phi^2$, and cannot reach $\beta_1 < 0$. Current P05 defines $\chi = 4(H-G)/C$ without a positive-sign restriction (lines 58–61) and explicitly covers all real signs in lines 304–319. Thus P07's all-sign threshold is legitimate: $H \leq G$ gives $\chi \leq 0 < \alpha_0$ and uniform selection. The reciprocal-vertex argument is restricted to $H > G$ exactly as required.

For physical parameters, $D_5 = x(h - g/s^2)$ and $C = c/s$ yield

$$h - \frac{g}{s^2} = \frac{c}{\phi^2 s x}, \qquad \boxed{h = \frac{g}{(1+x)^2} + \frac{c}{\phi^2(1+x)x}.}$$

All divisions use strictly positive x, s, ϕ. The equilibrium bracket must hold simultaneously. At the specified equilibrium, smaller h than the displayed right side selects class 0, equality ties classes 0 and 1, and larger h selects pair 1. This is a branch-state condition; it does not fix x independently while varying a physical tuple with all other entries held fixed.

At $T = C$, substitution in the gaps gives

$$r_1 = r_2, \qquad r_0 - r_1 = r_0 - r_2 = 5C/2 > 0.$$

At $T = -\phi^2 C$, the same calculation gives

$$r_0 = r_2, \qquad r_1 - r_0 = r_1 - r_2 = 5\phi C/2 > 0.$$

For example, the latter follows from $\sqrt{5}(1+\phi^2) = 5\phi$. Both are strictly subdominant ties, not additional mode-selection switches. Neither alone asserts equality of complex eigenvalues.

6 Frequency degeneracy, intersection, and local codimension

The independently reconstructed imaginary parts are

$$\omega_1 = \sigma(S - \phi C), \qquad \omega_2 = -\sigma(C + \phi S) < 0.$$

Every factor in the last strict inequality is positive. Hence labels 2 and 3 are always nonreal and distinct. The pair $\{1, 4\}$ is real exactly at $S = \phi C$, where $\lambda_1 = \lambda_4 = r_1$. P04's unitary diagonalization supplies semisimplicity even at repeated values. This condition by itself is no assertion of a zero eigenvalue, steady-state bifurcation, Hopf bifurcation, or first loss of stability.

Intersecting $H - G = C\phi^{-2}$ with $G + H = \phi C$ gives

$$\frac{G}{C} = \frac{\phi - \phi^{-2}}{2} = \phi^{-1}, \qquad \frac{H}{C} = \frac{\phi + \phi^{-2}}{2} = 1.$$

Both ratios are positive. For $F_1 = H - G - C\phi^{-2}$ and $F_2 = G + H - \phi C$, their full gradients in (C, G, H) are

$$(-\phi^{-2}, -1, 1), \qquad (-\phi, 1, 1).$$

The (G, H) minor is $\det \left(\begin{smallmatrix} -1 & 1 \\ 1 & 1 \end{smallmatrix}\right) = -2$. The common zero set is a nonempty codimension-two subset of the positive three-dimensional effective-coefficient domain.

To transfer this claim to physical parameters requires more. P05's verified equilibrium chart is the map

$$\Phi(x, k, c, g, h) = \left((k + h)x - \frac{c - g}{1 + x},\ k, c, g, h\right)$$

on the open set where x, k, c, g, h and its first component a are positive. Its determinant is

$$\det D\Phi = k + h + \frac{c - g}{s^2} = k + (h - g/s^2) + c/s^2.$$

At the intersection $H - G = C\phi^{-2} > 0$, so $h - g/s^2 > 0$ and this determinant is strictly positive. The inverse function theorem therefore identifies local branch coordinates with local physical coordinates on the specified positive branch.

In those equilibrium coordinates, hold x, k, c fixed and differentiate the *unnormalized* constraints F_1, F_2 with respect to (g, h):

$$\begin{pmatrix} -x/s^2 & x \\ x/s^2 & x \end{pmatrix}, \qquad \det = -2x^2/s^2 \neq 0.$$

They have rank two on the five-dimensional equilibrium chart, and that rank transfers under the local diffeomorphism. The resulting set is locally codimension two in physical parameter space on that branch. Varying g, h in the chart also varies a as prescribed by Φ; this is not a differentiation while keeping the whole physical tuple fixed. P07 makes no global projection, unfolding or path-crossing claim.

The canonical nonemptiness witness has

$$x = 1, \qquad (a, k, c, g, h) = (\sqrt{5}, 1, 2, 4/\phi, 1).$$

All entries are positive. In the *original* DM equation each component equals

$$\sqrt{5} - 1 + \frac{2}{2} - \frac{4/\phi}{2} - 1 = \sqrt{5} - 1 - \frac{2}{\phi} = 0,$$

because $1 + 2/\phi = \sqrt{5}$. Also $C = 1$, $G = 1/\phi$, $H = 1$, so $H - G = \phi^{-2}$ and $G + H = \phi$. The witness is therefore a valid canonical equilibrium satisfying both spectral conditions.

7 Triple value and exhaustive numerical-coincidence proof

At the intersection, $\omega_1 = 0$ and $r_0 = r_1$. Since $1 - \phi^{-2} = \phi^{-1}$,

$$\begin{aligned}\lambda_0 = \lambda_1 = \lambda_4 &= d + C - C\phi^{-2} = d + C/\phi \\ &= -kx - C - Cx/s + C/\phi \\ &= -kx - Cx/s - C\phi^{-2} < 0.\end{aligned}$$

Every summand in the final negative expression is strictly negative. The ordering proof gives $r_2 < r_0$, while $\omega_2 < 0$ ensures labels 2 and 3 are distinct from each other and from this real value. Exactly labels 0, 1 and 4 contribute to it. By P04 the corresponding three orthogonal Fourier coordinates give algebraic multiplicity three and geometric multiplicity three over $\mathbb{C}$. Over $\mathbb{R}$, the uniform vector and the real and imaginary parts of the pair-1 vector likewise give a three-dimensional real eigenspace. "Triple value" does not mean a Jordan block. The whole matrix is Hurwitz there.

There are ten unordered pairs of distinct labels. The following partition checks every one:

1. The pair $(1, 4)$ coincides exactly at $S = \phi C$.
2. The pair $(2, 3)$ never coincides because $\omega_2 < 0$.
3. Neither $(0, 2)$ nor $(0, 3)$ coincides: label 0 is real and labels 2 and 3 are nonreal.
4. A coincidence for $(0, 1)$ or $(0, 4)$ requires both $r_0 = r_1$ and $\omega_1 = 0$. These are precisely $T = -C\phi^{-2}$ and $S = \phi C$, the intersection already checked.
5. The four cross-pair choices $(1, 2), (1, 3), (4, 2), (4, 3)$ require $r_1 = r_2$, hence $T = C$. Then $G = H + C$ and $S = C + 2H > C$. Direct subtraction gives

$$\begin{aligned}\omega_1 - \omega_2 &= \sigma[(1 + \phi)S - (\phi - 1)C] > 2\sigma C > 0, \\ -\omega_2 - \omega_1 &= \sigma[(1 + \phi)C + (\phi - 1)S] > 0.\end{aligned}$$

   Thus $\omega_2 < \omega_1 < -\omega_2$, excluding both signs and, by conjugation, all four cross-pair equalities.

An equivalent adversarial check solves the two putative frequency equalities: they require respectively $S/C = \phi^{-3} < 1$ or $S/C = -\phi^3 < 0$, both incompatible with $S/C > 1$ at $T = C$. This is an analytic obstruction, not a failure of a numerical search.

Consequently P07's three cases are exhaustive:

Conditions	Only numerical repetition
$S \neq \phi C$	None; all five values are simple.
$S = \phi C$, $T \neq -C\phi^{-2}$	$\lambda_1 = \lambda_4$; multiplicity two.
$S = \phi C$, $T = -C\phi^{-2}$	$\lambda_0 = \lambda_1 = \lambda_4$; multiplicity three.

All listed multiplicities are both algebraic and geometric. Pair 2 always consists of simple numerical eigenvalues sharing no value with another label. Every nonreal pair-1 eigenvalue is simple.

On the frequency locus, strict $G, H > 0$ imply $|G - H| < G + H = \phi C$, hence $-\phi < t < \phi$. The value $t = 1$ is allowed: explicitly $G/C = (\phi + 1)/2$ and $H/C = (\phi - 1)/2$ are positive. Here $r_1 = r_2$ but pair 1 is real and pair 2 is nonreal, so their numerical values differ. In contrast $t = -\phi^2$ is outside $(-\phi, \phi)$ since $\phi^2 > \phi$.

8 Hurwitz criterion, stable dominant tie, and critical-mode ledger

The P04 linear stability test is $\mathrm{sa}(J) = \max(r_0, r_1, r_2) < 0$. The independently proved ordering reduces it, including equality, to

$$\begin{cases} r_0 < 0, & t > -\phi^{-2}, \\ r_0 = r_1 < 0, & t = -\phi^{-2}, \\ r_1 < 0, & t < -\phi^{-2}. \end{cases}$$

The middle case tests the common value. It is automatically satisfied at every canonical positive equilibrium on that tie: using only $T = -C\phi^{-2}$ and the equilibrium expression for d,

$$\boxed{r_0 = r_1 = d + C - C\phi^{-2} = -kx - \frac{Cx}{1+x} - C\phi^{-2} < 0.}$$

No condition on S was used. This establishes the whole tie's strict stability independently of the frequency intersection. Since $r_2 < r_0$ there, the dominant switch is never itself a zero-spectral-abscissa surface in the positive canonical domain. This is only the spectral consequence stated by P07, not a bifurcation classification.

Normality also verifies the source's boundary language. If the maximum is zero, P04 gives $\|e^{\tau J}\|_2 = e^{\tau\,\mathrm{sa}(J)} = 1$ for $\tau \geq 0$; the linear system is Lyapunov stable and not asymptotically stable. A positive maximum is linearly unstable. P07 does not infer nonlinear boundary stability from either statement.

Class 0 attains the maximum exactly when $t \geq -\phi^{-2}$; pair $\{1, 4\}$ does so exactly when $t \leq -\phi^{-2}$; pair $\{2, 3\}$ never does. At a zero-abscissa state the first two eligibility inequalities must be strict, since the shared boundary is strictly stable. If $r_2 = 0$, the direct exclusion proof supplies either $r_1 > r_2 = 0$ or $r_0 > r_2 = 0$. Thus pair 2 cannot be a sole or tied controlling class of a stability boundary. These are fixed-state facts, with no assumption of prior stability or an oriented path.

9 Canonical realizability and exact downstream-input review

For arbitrary $C, G, H > 0$ take

$$x = 1, \quad k = C, \quad c = 2C, \quad g = 4G, \quad h = H, \quad a = H + 2G.$$

All five physical coefficients are strictly positive. The original five-term component, without relying on an already simplified root formula, is

$$(H + 2G) - C + \frac{2C}{2} - \frac{4G}{2} - H = 0.$$

The effective coefficients are $c/2 = C$, $g/4 = G$, $h = H$, and the original diagonal derivative equals

$$H + 2G - 2C - \frac{2C}{4} - \frac{4G}{2} - H = -\frac{5C}{2}.$$

For any real t, let $C = 1$, $G = 1 + \max(t, 0)$ and $H = 1 + \max(-t, 0)$. Both latter coefficients are at least 1 and $G - H = t$ in both sign cases, including $t = 0$. This construction realizes every strict interval and every equality row in the ordering table with a genuine canonical positive equilibrium. It does not assert that arbitrary d is realizable independently, and it uses no MB theorem.

Each claimed P11 input in P07 **L494–508** is checked separately:

1. The only possible nonuniform dominant class is $\{1, 4\}$: the two-case gap proof excludes pair 2 everywhere.
2. Its signed imaginary part is $\sigma(S - \phi C)$ with the orientation inherited from the specified Fourier convention.
3. Its nonzero-frequency condition is exactly $S \neq \phi C$, since $\sigma > 0$.
4. If pair 1 maximizes, $t \leq -\phi^{-2} < 1$, so $r_1 - r_2 = \sqrt{5}C(1 - t)/2 > 0$, including the dominant tie. If pair 1 is uniquely maximal, the second gap also gives $r_0 < r_1$.
5. Every nonreal pair-1 numerical eigenvalue is simple, by the exhaustive coincidence proof and P04's multiplicity rule.
6. No pair-2 label numerically coincides with any other label, including its own conjugate label.
7. The only uniform/pair-1 numerical coincidence is the strictly negative semisimple triple value. More generally the entire real-part dominant tie is strictly stable.

These checks add no P11 conclusion. Frequency and simplicity alone do not establish a simple Hopf pair along a path, a first instability, transversality or nonlinear nondegeneracy.

10 Adversarial search and independent computational evidence

The attempted counterexamples have analytic resolutions:

Target to break	Reason it cannot fail in the stated domain
Pair 2 never dominant	Split at $t = 1$: one of $r_1 - r_2, r_0 - r_2$ is strictly positive.
Distinct values off $S = \phi C$	All ten unordered label pairs are exhausted; uniform/pair-1 equality forces the frequency locus.
No cross-pair equality	Real equality forces $T = C$ and $S > C$; the two signed frequency differences are strictly positive.
Entire dominant tie stable	The equilibrium identity produces $-kx - Cx/(1 + x) - C\phi^{-2}$ without using frequency degeneracy.
Only labels 0,1,4 at the intersection	Pair 2 has smaller real part and strictly nonzero opposite imaginary parts.
Every ordering cell realizable	A positive original-DM equilibrium is constructed for every real t; equality rows included.

New independent checker. `A07_support/independent_check.py` was written without importing or copying the developer checker, after the analytic reconstruction. It uses rational arithmetic in $\mathbb{Q}(\sqrt{5})$ and the extension $r + \mathrm{i}\sigma w$, with $r, w \in \mathbb{Q}(\sqrt{5})$ and $\sigma^2 = (5 - \sqrt{5})/8$. Rational sign decisions use squared comparisons only. Positive radical choices are justified in Section 3; they are not inferred from floating-point values.

The run used Python 3.12.14 and NumPy 2.3.5 and passed **953 exact assertions**, **91 exact positive coefficient cases**, and **483 numerical canonical matrix cases**. The exact suite includes coefficient-basis identities for the five eigenvalues and all gaps, all phase rows, all seven orderings at three scales,

six frequency-locus settings, a 4^3 positive rational coefficient grid, exhaustive ten-pair coincidence tests, the intersection witness, and stable-tie/branch checks at five positive state values. There are **21 exact direct matrix cases**, each testing all five Fourier vectors in the original-component Jacobian. General inequalities and codimension arguments are proved in the audit, not inferred from this finite set.

The numerical sample consists of 63 ordering cases, 54 frequency cases, 6 nearby-threshold cases, 300 arbitrary positive examples and 60 targeted dominant-tie examples. The random seed is 70032026; the arbitrary samples range over six orders of magnitude in C, G, H, x. Each is converted to positive physical coefficients by $c = C(1+x)$, $g = G(1+x)^2/x$, $h = H/x$, $k = 1 + C/x$, and the equilibrium value of a. In particular $a = x + H + G(1+x)/x > 0$.

For every numerical case, the original vector field is evaluated at $x\mathbf{1}$. Its direct Jacobian entries are compared with complex-step derivatives of that original field. Direct matrix eigenvalues are compared to the analytic spectrum using all $5! = 120$ bijections; no set conversion or sorting discards multiplicity. Labelled Fourier residuals and the original exponential symbol are separate checks. The scaled tolerance is 2×10^{-10}. Largest errors were

Check	Maximum scaled error
Original equilibrium residual	2.14×10^{-16} (upper rounded)
Matrix versus complex-step derivative	1.66×10^{-13} (upper rounded)
Unordered matrix spectrum, multiplicities retained	2.61×10^{-15} (upper rounded)
Labelled Fourier residual	4.01×10^{-16} (upper rounded)
Original exponential symbol	5.99×10^{-16} (upper rounded)

The complete inputs, spectra and errors are retained in the exact and numerical result files. No counterexample was found. This observation is not a proof of a universal claim, and floating tolerances do not certify a coincidence or a strict inequality arbitrarily near a locus.

Separate developer-validation provenance. The previously named `P07_validation_support.zip` was located in the existing P07 delivery. Its SHA-256 is

`cabf306b6a95db0951f320ba472ee401f050ec014a7ee12b6a558f9f41b2abc5`

Its recorded draft archive hash and all 118 current-member hashes match the supplied project. After the independent derivation and checker were completed, the developer checker was inspected and rerun in an isolated audit-support output directory. Its 161 exact checks and 108 numerical cases pass. The reproduced summary, exact-results and numerical-results files are each byte-identical to the originals; the comparison and immutable checker copy are retained in `A07_support/prior_validation/`. Thus there is no new unresolved P07 validation-provenance issue. This reproduction does not supply a proof premise, make the checks independent, or confer project status.

11 Line-by-line source review

The following contiguous ranges partition all 555 source lines, including blank lines and comments. Each substantive assertion within each range was read and checked. The allowed review labels are PASS, MINOR ISSUE, MAJOR ISSUE, FALSE and UNRESOLVED; only PASS is needed for this source. Specific evidence is given above and summarized here. No intended correction has been silently substituted for the literal current text.

P07 lines	Verdict	Finding and independent check
1–16	PASS	Metadata correctly identifies DRAFT, inherited/direct prerequisites, exact version and spectral-only scope. P08 is expressly excluded as theorem input.
17–24	PASS	Section/status and dependency prose agree with current manifest. P05 → P07 is preserved.
25–41	PASS	Original DM orientation, strict coefficient signs, local s, S, D_5 definitions and signed-frequency terminology are correct.
42–48	PASS	Original equilibrium bracket and $d = -kx - C - Cx/s$ follow by direct substitution; equilibrium use is explicitly limited.
49–73	PASS	All four golden-ratio identities, both cosines, their difference and both positive sine radicals are exact.
74–87	PASS	Fifth-root geometric sum and cosine quadratic select the correct root; positive sine signs fix the radicals.
88–103	PASS	Every phase row has correct quadrant signs, especially $\sin 2\theta_2 = -\phi\sigma$ and $\sin 2\theta_3 = \phi\sigma$.
104–109	PASS	Three representative classes and their five labels are counted correctly; no odd-dimensional Nyquist singleton.
110–132	PASS	All five labelled eigenvalues and the three real-part classes follow from the reconstructed phases; conjugate multiplicities retained.
133–149	PASS	P04 import is exact. The $-H$ shift has negative real and positive imaginary sine contribution.
150–156	PASS	P05 quadratic checks both nonuniform classes, using both double-cosine identities.
157–170	PASS	All three gap factors and powers of ϕ are correct; $C > 0$ legitimizes $t = T/C$.
171–189	PASS	Seven cells exhaust the real line; both subdominant ties and the unique dominant tie are correctly distinguished; d cancels.
190–209	PASS	Gap coefficients, positive prefactors, seven sign patterns and absence of a triple class tie all check; no independent physical d is inferred.
210–230	PASS	Sign reversal $D_5/C = -t$ is handled correctly; selected representatives and full label sets include equality.
231–238	PASS	Direct split $t < 1$ versus $t \geq 1$ excludes pair 2 even from tied dominance.
239–250	PASS	$\beta_0 = \phi^2/4$, $\beta_1 = -1/4$, $\alpha_0 = 6 - 2\sqrt{5}$; positive vertex meets only the first midpoint.
251–266	PASS	Matches current P05's all-sign χ definition. $H \leq G$ is correctly uniform; reciprocal argument is confined to $H > G$.
267–281	PASS	Canonical switch has every denominator correct and retains the simultaneous equilibrium equation; branch-state interpretation and P05 regularity apply.
282–290	PASS	Subdominant gaps are $5C/2$ and $5\phi C/2$. Neither real-part tie is declared a numerical coincidence.
291–308	PASS	Both signed frequencies, strict $\omega_2 < 0$, exact pair-1 real locus and semisimplicity statement are valid.
309–317	PASS	Proof uses positive factors, conjugacy and P04's repeated-value multiplicity theorem; no bifurcation inference.
318–336	PASS	Intersection ratios, nonempty local codimension-two statement, strictly negative triple value, multiplicity three and Hurwitz conclusion hold.
337–345	PASS	Sum/difference solution is positive; effective (G, H) gradient determinant is exactly -2.
346–353	PASS	Original DM witness residual is zero; $C = 1, G = \phi^{-1}, H = 1$; all physical parameters positive.
354–362	PASS	P05's chart determinant is positive on $H - G > 0$; physical constraint minor is $-2x^2/s^2$. Local physical transfer is justified.
363–370	PASS	Triple-value derivation uses equilibrium correctly; exactly three independent Fourier vectors, with pair 2 distinct and strictly below.
371–386	PASS	All three numerical-coincidence cases are exhaustive; simplicity and algebraic/geometric multiplicities use numerical values, not merely label count.

P07 lines	Verdict	Finding and independent check
387–393	PASS	Within-pair and uniform-versus-pair possibilities account for six unordered label pairs and force exactly the stated intersection when applicable.
394–408	PASS	Remaining four cross-pair equalities require $T = C$, hence $S > C$; both frequency inequalities have correct signs and are strict.
409–414	PASS	$\lvert T\rvert < S$ gives $-\phi < t < \phi$; $t = 1$ is realizable there, $t = -\phi^2$ is not; real parts and values remain distinct concepts.
415–435	PASS	Piecewise Hurwitz criterion has correct strict inequalities and equality row. Entire dominant tie has strictly negative common maximum.
436–441	PASS	Stable-tie proof uses $T = -C\phi^{-2}$ and equilibrium d only, independently of $S = \phi C$.
442–448	PASS	Zero maximum implies linear Lyapunov stability, not asymptotic stability, by P04 normality. No nonlinear or bifurcation claim; dominant tie excluded from zero abscissa.
449–469	PASS	Fixed-state eligibility table, strict zero-abscissa restrictions and exclusion of pair 2 as controlling class are correct; no first-path claim.
470–476	PASS	At every point some other real part strictly exceeds r_2; when $r_2 = 0$ this other real part is positive.
477–485	PASS	Every positive effective triple is realized with $x = 1$ and positive physical parameters; $d = -5C/2$ exactly.
486–493	PASS	Original residual vanishes; the max-function construction has $G - H = t$ for every real t and makes all seven rows nonempty.
494–508	PASS	All seven downstream spectral inputs and strict gaps are valid. Hopf existence, first crossing, derivatives and nonlinear conclusions are explicitly deferred.
509–518	PASS	Fifth-root origin is correctly delimited; no physiological law, optimum, medical claim, empirical TCM validation or standalone novelty assertion.
519–527	PASS	Named companion located and authenticated; its coverage checked and results reproduced separately. Source explicitly denies proof/status from computation.
528–540	PASS	Auditor checklist accurately identifies phase, gap, equality, sign-convention and equilibrium checks without asserting new results.
541–550	PASS	Checklist correctly separates codimension, multiplicity, coincidence, whole-tie stability and fixed-state eligibility from bifurcation claims.
551–555	PASS	Current prerequisite authentication, direct edge, DRAFT lifecycle and independence requirements are consistent with the actual audit boundary.

12 Scope, corrections, and closure handoff

The dependency scan and the proof reconstruction agree: P07 uses only the canonical contract and notation, inherited P01/P02 foundations, verified P03/P04/P05 results, and elementary exact algebra and trigonometry. References to P08 are version/scope context and the local-symbol disambiguation at **L37–41**; no P08 theorem is invoked. The preserved mathematical edge is P05 → P07. The frozen scheduling edge P08 then P07 creates no extra theorem dependency.

The literal uses of "all", "only", "never", "simple", "multiplicity", "codimension two", "Hurwitz", "critical mode" and "first instability" were checked in their surrounding sentences. Their quantifiers and qualifications are correct. Source occurrences of Hopf, transversality, periodic orbit and rotating wave are scope exclusions or future-input descriptions, not accidental positive claims. The golden ratio is fifth-root Fourier geometry only; the rejected C155 interpretation stays rejected. The frozen priority limitations remain unchanged, and no literature survey was undertaken.

Required correction ledger: empty. No minor or major issue, false theorem, unresolved mathematical claim or new unresolved P07 validation-provenance issue was found. Existing A04/A05 historical

notes are neither erased nor promoted into new mathematical objections. All five layer verdicts and the overall verdict are VERIFIED for the exact source hash specified in Section 1.

Closure handoff. The original development chat may record this exact-version A07 result and perform its normal administrative closure. No mathematical or wording change to P07 is required by A07. If the mathematical body is changed before closure, this audit should not be silently applied to the changed content. After closure, P09 is cleared under the frozen attempt order; this is readiness, not a P09 result or an instruction to begin it during this audit.

Audit-only delivery. The package is named exactly `cyclic_network_project_A07_audit_only.zip`, rooted at `cyclic_network_project/`, and contains this standalone A07 TeX source, its PDF reading copy, exact audited-source/dependency hashes, the complete line review, independent check scripts/results and separate validation-provenance records. It contains no replacement P07 source and no full-project snapshot. The supplied draft archive and all project proof bytes remain unchanged. No archive named `cyclic_network_project_P07_audited.zip` is produced by this task.